5.3.1 Using tables in individual steps

Suppose you're working on a feature that allows Frequent Flyer members to transfer points to other members. For example, suppose Danielle and Martin are Frequent Flyer members. Both Danielle and Martin have accumulated a lot of points over the year. They want to go on holidays together using their points, but neither of them has enough points to buy the tickets outright. Martin needs to be able to transfer some of his points to Danielle so that she can purchase the flights for both of them with her points.

You could express this scenario as follows:

```
Scenario: Transfer points between members
Given Danielle's account has 100000 points and 800 status points
And Martin's account has 50000 points and 50 status points
When Martin transfers 40000 points to Danielle
Then Martin should have 10000 points and 50 status points
And Danielle should have 140000 points and 800 status points
```

A lot of duplication here

Here too

The problem is that there's a lot of repetition and clutter in this scenario, and the meaning gets lost in all the words. A much better way to write this scenario would be to express the data in a more concise tabular format, like this:

```
Scenario: Transfer points between existing members
    Given the following accounts:
      | owner     | points | status-points |
      | Danielle  | 100000 | 800           |
      | Martin    | 50000  | 50            |
    When Martin transfers 40000 points to Danielle
    Then the accounts should be the following:
      | owner    | points | status-points  |
      | Danielle | 140000 | 800            |
      | Martin   | 10000  | 50             |
```

❶ **Provide table of data for the Given step**

❷ **Data for the Then step**

The tables start directly after the `Given` ❶ and `Then` ❷ steps, with the values being separated by pipes (`|`). The headers at the top of each column are useful in this case, but are optional. For example, in the following step, you could provide a list of values:

```
Then I should be able to upgrade to one of the following cabin classes:
| Premium Economy |
| Business        |
```

Embedding tabular data is a great way to express preconditions and expected outcomes in a clear and concise way. But there's another equally useful way to use tabular data in your scenarios: tables of examples.

5.3.2 Using tables of examples

Suppose you need to implement the feature that calculates the bonus points that travellers earn based on their Frequent Flyer status—the feature that Sarah and Paul discussed in section 5.1. The basic business rules they discussed were the following: